Indian IPO Analysis

A Comprehensive Quantitative Study of Listing Returns, Predictive Modelling, and Investment Strategy

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Research Question

Is investing in Indian IPOs a profitable strategy, and which factors most reliably predict IPO success across the listing day and subsequent holding periods?

Datasets: Capital Market (scraped), InvestorGain GMP (API), Yahoo Finance (enrichment)
Coverage: 1999–2026 | Mainboard (1,152 analysed IPOs) | SME (1,463 analysed IPOs) | GMP (~1,075 IPOs)

Abstract

This report presents a comprehensive quantitative analysis of the Indian Initial Public Offering (IPO) market using three independently constructed datasets spanning 1999–2026, covering 1,152 analysed mainboard IPOs, 1,463 analysed SME IPOs, and 1,075 GMP-era IPOs enriched with Grey Market Premium data, Yahoo Finance fundamentals, and post-listing return series at seven horizons. We evaluate five investment strategies of increasing sophistication—from the naive all-IPO baseline to a machine-learning-based classifier— and rigorously test each using appropriate nonparametric statistical methods.

Key findings are: (i) the all-IPO strategy produces statistically significant positive expected returns (mainboard mean 26.05%, median 9.96%, win rate 72.6%), but the mean overstates the typical experience given extreme right-skewness; (ii) subscription filtering is the most reliable mechanical screen, with the $> 100\times$ bucket achieving a 100% historical win rate; (iii) listing-day flipping dominates all holding horizons up to two years on a risk-adjusted basis (paired t-test, $p = 0.377$); (iv) Grey Market Premium is the single strongest predictor of listing gain (Spearman $\rho = 0.743$, $R^2 = 0.697$, slope ≈ 1.0); and (v) chronological ML validation produces holdout ROC-AUCs around 0.82–0.86, with a train-CV-selected screening threshold lifting the holdout win rate to 92.5% versus a 55.3% baseline.

Contents

Notation and Abbreviations

Symbol / Abbreviation	Meaning
μ	Population mean listing gain
$\hat{p}_q$	q-th sample quantile
ρ	Spearman rank correlation coefficient
r	Pearson correlation coefficient
R^2	Coefficient of determination (OLS)
η^2	Effect size (Kruskal-Wallis equivalent)
GMP	Grey Market Premium
GMP%	GMP as percentage of offer price
QIB	Qualified Institutional Buyer
NII / HNI	Non-Institutional Investor / High Net-worth Individual
RII	Retail Individual Investor
SME	Small and Medium Enterprise exchange segment
MB	Mainboard
PAT	Profit After Tax
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
AUC	Area Under the ROC Curve
OFS	Offer for Sale
DRHP	Draft Red Herring Prospectus
BRLM	Book Running Lead Manager

Abbreviations and notation used throughout this report

1 Introduction

1.1 Background and Motivation

An Initial Public Offering (IPO) is the process by which a private company raises capital from the public by issuing shares on a stock exchange for the first time. In India, the IPO market has undergone a remarkable transformation over the past two decades, evolving from a relatively niche vehicle for public-sector enterprises into one of the most actively traded segments of the capital markets. The period from 2020 onwards, in particular, witnessed an unprecedented boom in new listings, with 2024 recording 257 mainboard IPOs and 2025 setting an all-time record of 322 IPOs.

For retail investors in India, IPO investing carries a distinct cultural significance. The possibility of listing-day gains has driven millions of Demat accounts to be opened with the sole aim of applying for every new offering. Yet, the empirical evidence on whether this strategy works—and under what conditions—has remained largely anecdotal, fragmented, or confined to specific time windows.

This project undertakes a rigorous, data-driven investigation of the Indian IPO market using three independently constructed datasets spanning 1999 to 2026. It moves beyond simple return calculations to examine the mechanics of IPO pricing, the information content of pre-listing signals, the difference between mainboard and SME listings, and the feasibility of building machine-learning classifiers to screen investment opportunities.

1.2 Research Objectives

The project is structured around five interconnected research objectives:

1. **Establish the base rate.** Quantify the statistical properties of IPO listing-day returns across both the mainboard and SME segments, including win rates, distributional characteristics, and trend over time.
2. **Evaluate investment strategies.** Rigorously test a hierarchy of strategies from the naive "apply to every IPO" baseline through increasingly selective filters based on oversubscription, GMP, and fundamental quality.
3. **Determine the optimal holding period.** Investigate whether listing-day flipping or longer-term holding produces better risk-adjusted returns, using data at 1-week, 1-month, 3-month, 6-month, 1-year, 2-year, and 3-year horizons.
4. **Quantify predictive signals.** Assess the individual and combined predictive power of Grey Market Premium (GMP), total oversubscription, QIB participation, market phase, and macroeconomic momentum for listing-day outcomes.
5. **Build and validate machine-learning models.** Train binary classifiers and regression models that use only pre-listing information to predict whether a given

IPO will produce a meaningful listing-day pop ($\geq 10\%$), and backtest their portfolio-level implications.

1.3 Report Organisation

The remainder of this report is structured as follows. Section 2 describes the data construction methodology, including the scraping pipeline and enrichment process. Section 3 covers data cleaning and preprocessing. Section 4 presents exploratory data analysis. Sections 5–9 analyse five investment strategies of increasing sophistication. Section 10 examines the effect of market regime on IPO performance. Section 11 compares the SME and mainboard segments. Section 12 studies loss-making company IPOs. Sections 13 and 14 present the machine-learning models. Section 15 presents the model-based strategy backtest. Section 16 covers sector-level analysis, and Section 19 presents conclusions and recommendations.

2 Data Construction and Methodology

2.1 Data Sources Evaluated

A systematic review of candidate data providers was conducted before settling on the final pipeline. The evaluation considered data completeness, historical depth, structured format, and availability without a paid subscription.

Table 1. Candidate data sources evaluated and their suitability

Source	Notes	Verdict
Chittorgarh	Previously free; historical data now behind the paid "IPO Matrix" product	Rejected
IPO Matrix	Structured historical data; trial version limited to 2 recent years	Rejected (cost)
NSE / Moneycontrol	Partial IPO data; requires significant scraping and cleaning	Supplementary
Capital Market	Uniform historical IPO table, stateful pagination, 2010–2026	**Primary (scraped)**
Kaggle (2010–2025)	Independent dataset used for spot-check validation only	Validation
InvestorGain	REST API for Grey Market Premium data, 2019–2026	GMP dataset
Yahoo Finance	Fundamental and price data via YahooQuery library	Enrichment

Capital Market was selected as the primary source because it provides a uniformly structured, paginated historical table with consistent column definitions across years—a feature absent in most alternatives.

2.2 Web Scraping Architecture

The Capital Market IPO historical table uses **stateful pagination**: the browser URL does not change when the user navigates between pages. The entire table interaction occurs through JavaScript-driven DOM manipulation, making standard HTTP-based scrapers ineffective. A browser-automation approach using `Selenium` with `ChromeDriver` was therefore required.

2.2.1 Pagination Logic

The pagination control of the Capital Market website renders three page numbers at a time in a sliding window. The correct traversal sequence is:

Algorithm 1 Capital Market Stateful Pagination Traversal

```
Load page 1 and extract table
Click page 2; wait for DOM change; extract table
Click page 3; wait for DOM change; extract table
Click Next; window slides to pages 4, 5, 6
repeat
    Extract current page table
    Click next visible page number if available
    Else click Next to slide the window forward
until table signature repeats (duplicate detection)
Concatenate all pages; export to CSV
```

2.2.2 Duplicate Detection

To prevent infinite loops when the pagination reaches its end, a *table signature* was computed from the first and last rows of each extracted table. If a signature appeared twice, scraping halted. This was critical because the pagination "Next" button can remain visible even on the final page.

2.2.3 Key Implementation Details

- **Element identification:** Pagination controls were identified by restricting search to <a> and <button> tags with numeric text content. This prevented issue sizes and prices (also numeric) from being mistaken for pagination buttons.

- **Click strategy:** Elements were scrolled into view and clicked via JavaScript (`execute_script`) rather than the Selenium `click()` method, which is fragile on elements behind overlays.
- **Wait strategy:** After each click, a custom `wait_until_table_changes()` function polled the page until the table signature changed, with a 30-second timeout.
- **Table parsing:** BeautifulSoup identified the target table by token-matching against known column headers (`List Date`, `IPO Name`, `Listing Gains`). `pandas.read_html()` converted the HTML to a DataFrame, with multi-level column headers flattened automatically.
- **Incremental saving:** Each page was saved to a separate CSV in a `pages/` subdirectory before proceeding, providing fault-tolerance against connection drops.

Capital Market IPO Historical Table: scraped target columns

List Date	IPO Name	Issue Size (Cr)	Offer Price	List Price	Listing Gains %
14 May 2026	Bagmane Prime Offi	3405.0	100.0	103.4	3.4
08 May 2026	Onemi Technology S	925.92	171.0	191.0	11.7
29 Apr 2026	Citius Transnet In	1105.0	100.0	104.5	4.5
17 Apr 2026	Om Power Transmiss	150.06	175.0	181.1	3.49
02 Apr 2026	Sai Parenteral's L	408.79	392.0	405.0	3.32
02 Apr 2026	Powerica Ltd	1100.0	395.0	375.0	-5.06
02 Apr 2026	Amir Chand Jagdish	440.0	212.0	195.0	-8.02
30 Mar 2026	Central Mine Plann	1841.45	172.0	162.8	-5.35

Figure 1. Capital Market IPO historical table (source: `capitalmarket.com`)

2.3 GMP Data Collection

Grey Market Premium data was collected from the InvestorGain REST API endpoint, which returns paginated JSON for each year from 2019 to 2026. The API URL template is:

```
https://webnodejs.investorgain.com/cloud/v2/report/data-read/377/{page}/5/{year}/...
```

Fields including GMP value, IPO price, listing price, closing price, subscription multiple, and IPO category (mainboard vs SME) were extracted and cleaned. HTML entities embedded in certain fields were decoded using Python's `html.unescape()` function and BeautifulSoup for residual markup. The final GMP dataset covers 2019–2026 with approximately 1,075 distinct IPOs after deduplication.

2.4 Dataset Enrichment

All three datasets (mainboard, SME, GMP) were enriched by appending additional fields from Yahoo Finance via the `yahooquery` Python library. The enrichment process used

Gemini AI for ticker symbol resolution (mapping IPO company names to NSE/BSE ticker codes), followed by API calls for:

- **Financial fundamentals:** Profit after tax (PAT), profit margins, return on equity, revenue growth, debt-to-equity ratio, trailing P/E, price-to-book.
- **Ownership:** Percentage held by insiders and institutions.
- **Post-listing returns:** Calculated at 1-week, 1-month, 3-month, 6-month, 1-year, 2-year, and 3-year horizons from listing date.
- **Risk metrics:** Maximum drawdown since listing, highest return since listing, beta.
- **Sector and industry:** Mapped from Yahoo Finance's classification.
- **Market regime features:** Nifty 50 return over 1M and 3M windows before each IPO's listing date, Nifty volatility, and the rolling 30/90-day average listing gain of the prior IPO cohort.

2.5 Final Dataset Summary

Table 2. Final datasets used in the analysis

Dataset	Coverage	IPOs	Primary Use
Mainboard Raw, deduplicated	1999–2026	1,152	Returns, subscription, strategy tests
SME Raw, deduplicated	2012–2026	1,463	SME-specific analysis
GMP Enriched	2019–2026	∼1,075	GMP prediction, ML models, long-run returns

The mainboard dataset includes columns for listing date, IPO name, issue size (INR crore), QIB/NII/RII/total subscription, offer price, listing price, listing gain percentage, and current market price. The GMP dataset additionally provides GMP amount and percentage, estimated listing price, sector, industry, market-phase labels, and all post-listing return horizons.

2.6 Validation Against Kaggle Reference Dataset

To validate the scraped mainboard data, a spot-check was conducted against a publicly available Kaggle dataset covering 2010–2025. Fifty IPOs were randomly sampled, matched by name and listing date, and compared on offer price, listing price, and listing gain percentage. Discrepancies exceeding ±1% were flagged for manual review. The validation confirmed high agreement between the two sources, establishing confidence in the scraping pipeline.

3 Data Cleaning and Preprocessing

3.1 Philosophy and Guiding Principles

The cleaning process was guided by four principles:

1. **Drop minimally.** Rows were discarded only when the target variable (listing gain) or the listing date was missing, as these are ineligible for analysis regardless.
2. **Impute with column medians.** Numeric features were imputed with the column median rather than the mean, as IPO distributions are strongly right-skewed and the median is a more robust central estimate.
3. **Flag informative missingness.** Where the absence of data is itself informative (e.g., no subscription breakdown for older fixed-price IPOs), a binary indicator column was created.
4. **Winsorise extreme outliers.** Listing gains were clipped at the 1st and 99th percentiles to prevent a small number of extreme performers from distorting statistical summaries and regression coefficients.

3.2 Mainboard Cleaning

Column names from the scraped CSV were remapped from Capital Market's verbose multi-level headers (e.g., `Price(Rs.) Listing Gains(%)`) to clean snake_case identifiers (`listing_gain_pct`). The listing date column was parsed with format `%d %b %Y`. A `has_sub_data` binary flag was created to distinguish IPOs with subscription breakdown from older ones without.

Winsorisation bounds (mainboard): $[\hat{p}_1, \hat{p}_{99}]$, where $\hat{p}_q$ denotes the q-th sample quantile. The raw maximum listing gain in the dataset is 4,940%, making winsorisation essential for any regression or mean-based summary.

3.3 SME Cleaning

SME IPO data from Capital Market uses a different date format (`DD-Mon-YYYY`) and issue size values sometimes contain embedded commas (e.g., `1,500`). These were handled at parse time. The SME dataset has no subscription breakdown by investor category (QIB/NII/RII), so these columns were set to `NaN` for consistency with the mainboard frame, and analyses requiring subscription breakdown were restricted to mainboard data.

3.4 GMP Dataset Parsing

The GMP dataset from InvestorGain required the most cleaning. GMP values were stored as strings with INR symbols and embedded percentage annotations (e.g., `Rs.120 (8%)`). A custom `strip_currency()` function used regex to extract the leading numeric value:

$$\texttt{strip\_currency}(s) = \text{Re-search}(r[-+]?\backslash d + \backslash.?\backslash d^*, s) \to \texttt{float}$$

Subscription multiples were stored as strings ending in `x` (e.g., `47.8x`) and parsed with a `strip_x()` function. GMP was also normalised as a percentage of the offer price to enable cross-IPO comparison:

$$\text{GMP\%}_i = \frac{\text{GMP value}_i}{\text{Offer price}_i} \times 100$$

3.5 Missing Value Analysis

Figure 2. Missing value pattern for key columns in the GMP enriched dataset

Key missingness observations:

- **Long-run returns (2Y, 3Y):** Necessarily sparse for recent IPOs; 3Y returns available for only ∼208 IPOs.
- **Sector and industry:** Present for ∼66% of GMP IPOs; Yahoo Finance does not always classify newly listed companies immediately.
- **Fundamental ratios** (P/E, debt-to-equity, etc.): 50–60% missing, concentrated in smaller and SME IPOs with limited Yahoo Finance coverage.
- **GMP and subscription:** Near-complete for post-2019 data.

4 Exploratory Data Analysis

4.1 Distribution of Listing-Day Returns

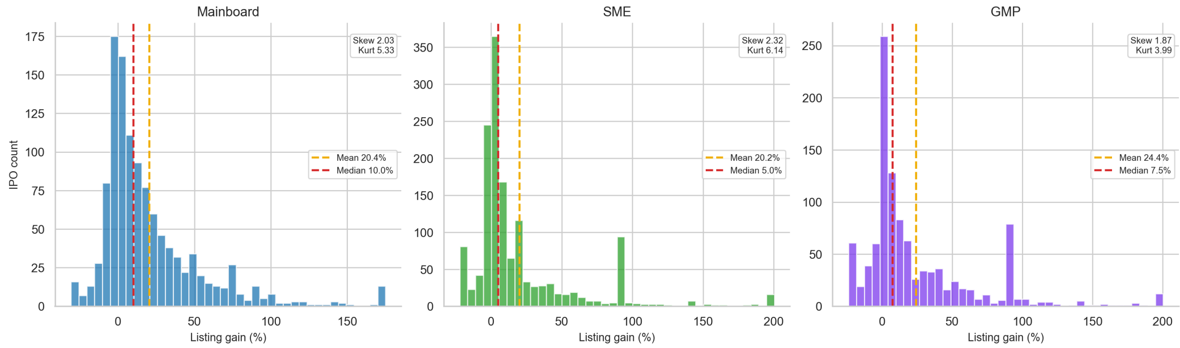

Figure 3. Listing-day gain distributions (winsorised at 1st–99th percentile)

The distributions reveal extreme right-skewness across all three datasets. Table 3 reports key distributional statistics.

Table 3. Distributional statistics for listing-day gains

Segment	N	Mean (%)	Median (%)	Std (%)	Win Rate	Skew	Kurt
Mainboard	1,152	26.05	9.96	152.41	72.6%	positive	high
SME	1,463	20.80	5.00	42.90	73.5%	positive	high
GMP Dataset	1,075	—	7.55	—	72.1%	positive	—

A robust inference check rejects the null of zero listing gain for both mainboard and SME data using bootstrap confidence intervals and nonparametric signed-rank tests. The t-test is retained only as a secondary diagnostic because IPO returns are heavy-tailed and strongly right-skewed.

Key insight: The *mean* gain substantially overstates the *typical* investor experience. The mainboard mean of 26.05% is driven by a small number of extreme performers. The median investor earns approximately 9.96% on mainboard and 5.00% on SME. This distinction has major practical implications for framing investment expectations.

4.2 IPO Issuance Volume Over Time

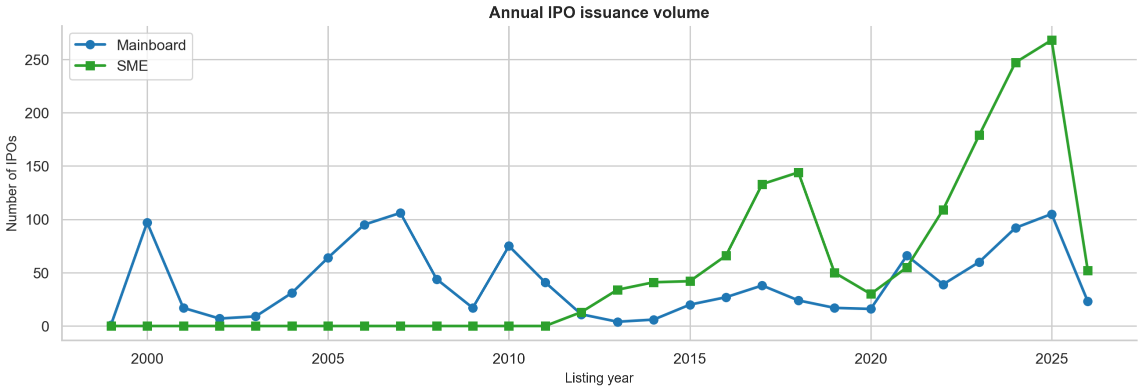

Figure 4. Annual IPO issuance: Mainboard and SME segments

IPO activity is strongly procyclical, tightly linked to Nifty 50 bull markets. Notable patterns include:

- **2019:** Only 4 GMP-era IPOs, median listing gain 1.52%, reflecting post-election uncertainty.
- **2021–2022:** Post-COVID liquidity boom; record retail participation.
- **2024:** 257 IPOs, median listing gain 30.36%, win rate 85.2% — the best single year in the dataset.
- **2025:** 322 IPOs (all-time record); median listing gain collapses to 3.17%, win rate 61.2% — supply saturation effect.
- **2026 (partial):** Median gain 0%, win rate 47.9% — the first year in the dataset where the typical investor generated zero return.

4.3 Issue Size Distribution

Mainboard issue sizes span four orders of magnitude, from sub-crore SME-scale offerings to mega-IPOs exceeding Rs.20,000 crore. A log-scale histogram is appropriate given this range. Larger issue sizes are associated with modestly lower listing gains (Spearman $\rho \approx -0.15$, $p < 0.01$), consistent with the intuition that larger offerings face greater supply-demand imbalance at listing.

4.4 Subscription Multiple vs Listing Gain

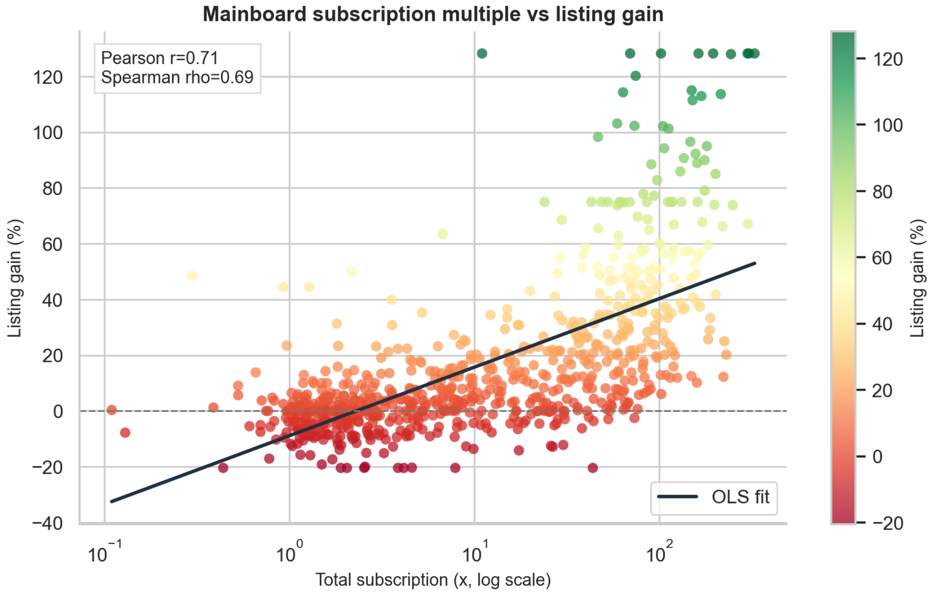

Figure 5. Total subscription vs listing gain — mainboard IPOs (OLS on log-subscription)

The scatter plot reveals a strong positive relationship between oversubscription and listing gain, with variance increasing at very high subscription levels (consistent with a multiplicative error model on log-subscription space). Correlation coefficients:

- Pearson $r = 0.56$ ($p < 0.0001$, on log-subscription)
- Spearman $\rho = 0.685$ ($p < 0.0001$)

The OLS slope in log-subscription space is approximately 12.3, meaning each unit increase in $\log(1 + \text{subscription})$ is associated with roughly 12.3 percentage points of additional listing gain.

4.5 Spearman Correlation Heatmap

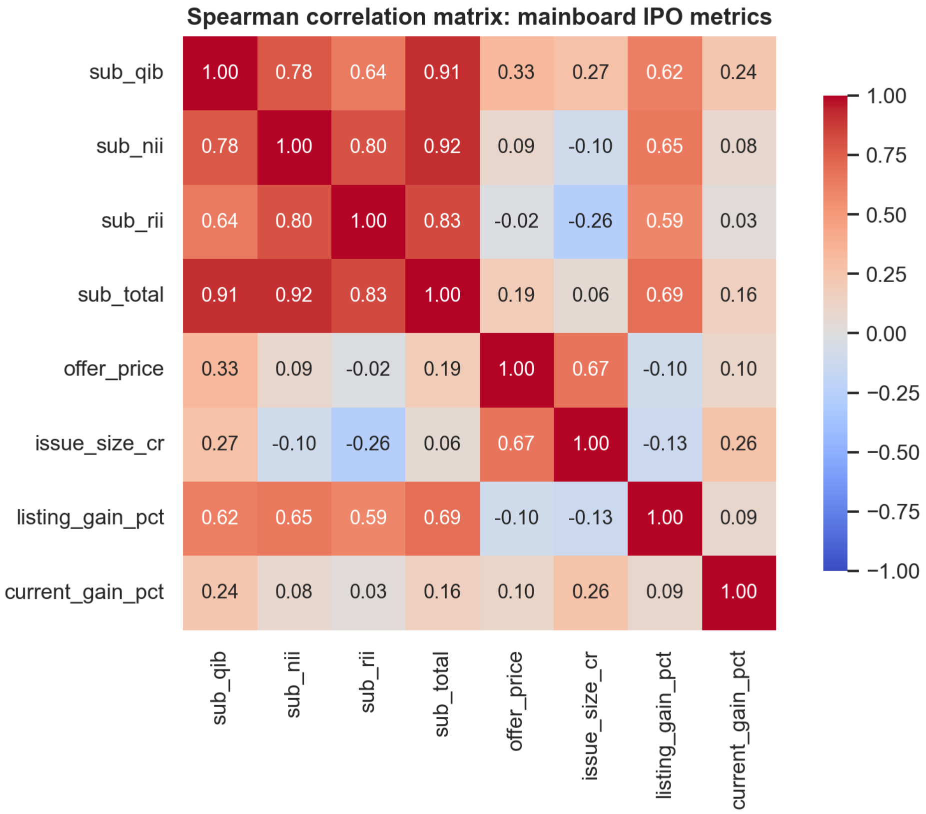

Figure 6. Spearman correlation matrix — mainboard IPO metrics

Key correlation findings from the heatmap:

- All four subscription categories (QIB, NII, RII, Total) are strongly positively correlated with each other ($\rho > 0.80$), creating multicollinearity. Including all four in a regression adds marginal information beyond the total.
- All subscription categories positively predict listing gain ($\rho = 0.58$–0.69).
- Issue size shows a modest negative correlation with listing gain: larger IPOs have more float and face less supply constraint at listing.

4.6 Year-over-Year Median Listing Gain

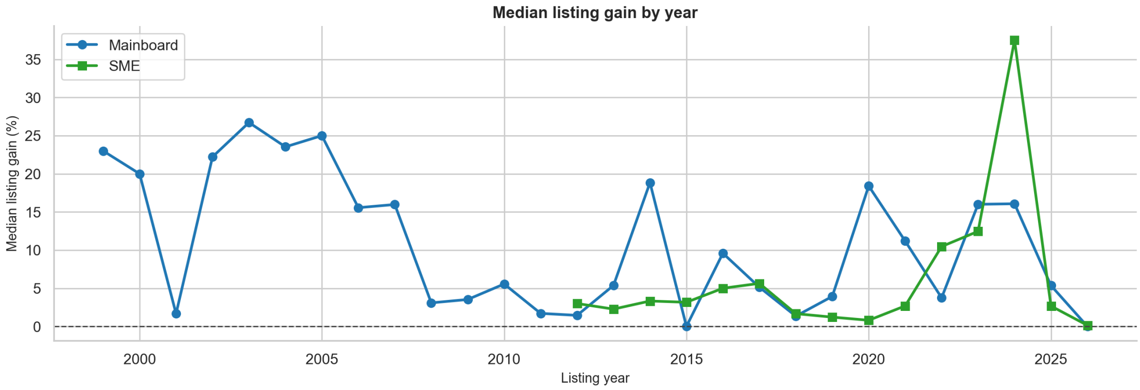

Figure 7. Median listing gain by year: Mainboard vs SME

5 Strategy 1: The All-IPO Baseline

5.1 Hypothesis

Applying for every IPO and selling at the listing-day close generates a positive expected return.

5.2 Methodology

For each of the mainboard, SME, and combined datasets, the following were computed:

- Win rate: proportion of IPOs with positive listing gain
- Mean and median listing gain
- Bootstrap 95% confidence interval for the mean (2,000 resamples)
- One-sample t-test: $H_0 : \mu = 0$

A portfolio simulation was also conducted: assuming Rs.10,000 of capital split equally across all IPOs in chronological order, selling at the listing-day close. This provides an upper-bound gross return benchmark, excluding allotment probability, brokerage, and taxes.

5.3 Results

Table 4. All-IPO baseline strategy results

Segment	N	Mean (%)	95% CI	Median (%)	Win Rate
Mainboard	1,152	26.05	[19.80, 36.26]	9.96	72.6%
SME	1,463	20.80	[18.68, 23.06]	5.00	73.5%
Combined	2,615	23.11	[20.08, 27.87]	6.49	73.1%

Bootstrap confidence intervals and signed-rank tests both support positive listing-day returns across the mainboard, SME, and combined universes. Mean-based t-tests are reported in the notebook as secondary checks, but the nonparametric results are more appropriate for these skewed distributions.

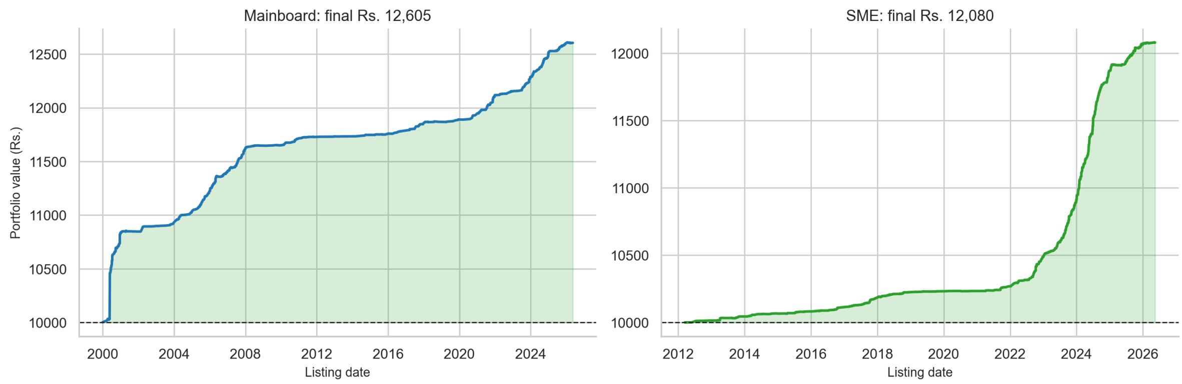

Figure 8. Equal-weight Rs.10,000 portfolio (sell at listing): Mainboard and SME

5.4 Interpretation

The all-IPO strategy works in expected-value terms but is uncomfortable in practice. With a 22.0% loss rate on mainboard, approximately 1 in 5 IPO applications results in a negative listing return. The strategy's variance (standard deviation $\approx$ 152.41%) is extremely high, driven by a small number of multi-bagger outliers. The median investor return of 9.96% on mainboard is real but unspectacular, especially after adjusting for the cost of capital and allotment probability (typically under 50% in oversubscribed IPOs).

Portfolio Equity Curve Interpretation

The Rs.10,000 equal-weight portfolio simulation (Figure 8) reveals an important structural pattern: mainboard gains are concentrated in the pre-2005 and post-2020 periods, with a long flat stretch from 2007–2019. This is consistent with the IPO cycle analysis in Section 4: the market was broadly flat or declining in IPO quality during the mid-cycle years. An investor who applied only during the 2020–2024 window would have earned dramatically higher returns than the long-run historical average implies.

For the SME portfolio, the curve is nearly flat from 2012–2021 and then steepens sharply in 2022–2024, confirming the SME segment's dependence on the post-COVID liquidity boom. This concentration of returns in a narrow time window makes the SME historical Sharpe-like ratio an unreliable guide to future performance.

6 Strategy 2: Selective Investment by Oversubscription

6.1 Hypothesis

Filtering IPOs by oversubscription level improves risk-adjusted returns over the all-IPO baseline.

6.2 Methodology

Five subscription-based filters were applied to the mainboard dataset (restricted to IPOs with subscription data available). For each filter, mean gain, median gain, win rate, standard deviation, and a raw Sharpe ratio (mean/std) were computed. A one-sided Mann-Whitney U test was used to test whether each filtered group stochastically dominates the full baseline ($\alpha = 0.05$).

6.3 Results

Table 5. Selective investment results by subscription filter — mainboard

Filter	N	Mean (%)	Median (%)	Win Rate	Std (%)	Sharpe*
All (baseline)	775	16.73	5.88	68.9%	37.47	0.447
Sub > median (∼20×)	387	32.13	21.62	88.6%	37.57	0.855
Sub > 75th pct	194	48.41	40.90	99.0%	35.27	1.373
QIB > 75th pct	194	47.27	37.47	98.5%	36.16	1.307
Sub > 50×	203	47.36	39.02	98.5%	36.39	1.302
Sub > 100×	81	63.90	56.67	**100.0%**	39.44	1.620

*Raw Sharpe = Mean/Std; no risk-free rate adjustment; single-period trades.

Statistical significance: Mann-Whitney U test rejects the null that the filtered group is no better than the baseline for all five filters ($p < 0.0001$ in each case).

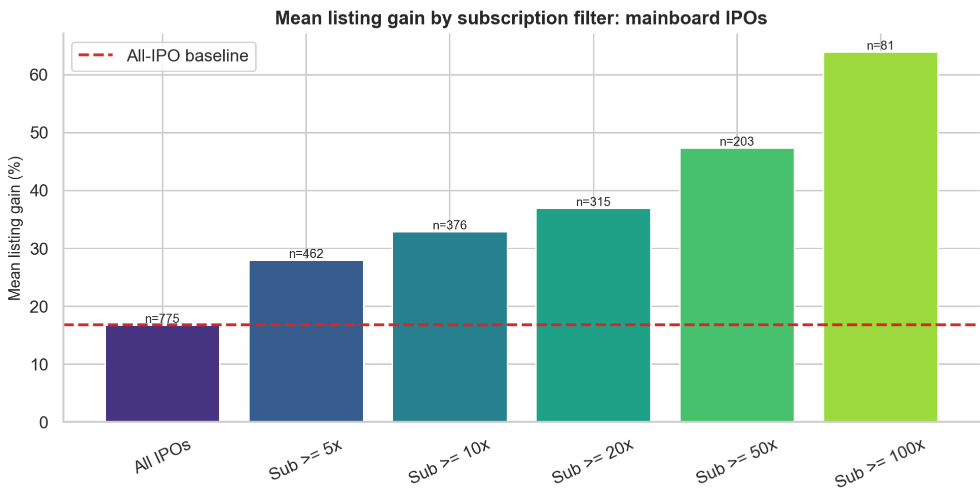

Figure 9. Mean listing gain by subscription filter — mainboard IPOs

6.4 Interpretation

The results show a near-monotone improvement in all performance metrics as the subscription filter becomes more stringent. The 100× subscription bucket is particularly striking: every historically observed mainboard IPO with total subscription above 100× produced a positive listing return ($N = 81$, win rate = 100%). The QIB > 75th percentile filter achieves 98.5% win rate and outperforms the total subscription filter at the same percentile, confirming that institutional due diligence is a marginally cleaner signal than aggregate retail enthusiasm.

Caution: These results are entirely in-sample and mechanically conditioned on completed IPOs. A 100% win rate over 81 historical observations does not guarantee future performance. In a prolonged bear market, even highly oversubscribed IPOs can list at discounts, especially if the offer price is aggressive or if the subscription multiple is inflated by leveraged NII demand. Treat the 100× rule as a strong screening signal, not as a deterministic trading rule.

7 Strategy 3: Optimal Holding Period

7.1 Hypothesis

Holding beyond the listing day yields better risk-adjusted returns than flipping at listing-day close.

7.2 Methodology

Post-listing returns were available from the GMP enriched dataset at seven horizons: 1 week, 1 month, 3 months, 6 months, 1 year, 2 years, and 3 years. Descriptive statistics (mean, median, win rate, Sharpe-like ratio) were computed for each horizon, with sample sizes decreasing for longer horizons (recent IPOs lack mature data). A paired t-test compared the 1-year return against the listing-day gain for the matched subset of IPOs where both were available ($N = 462$).

The Sharpe-like ratio for each holding period is computed as:

$$\text{Sharpe-like}_h = \frac{\bar{r}_h}{\sigma_h}$$

where $\bar{r}_h$ is the mean return at horizon h and σ_h is its standard deviation. This is *not* a time-adjusted Sharpe ratio: it does not annualise returns or subtract a risk-free rate. It is used purely as a within-sample signal-to-noise ratio to compare horizons.

7.3 Results

Table 6. Holding period analysis — GMP dataset

Period	N	Mean (%)	Median (%)	Win Rate	Std (%)	Sharpe-like
Listing Day	1,075	24.97	7.55	72.1%	45.11	0.554
1 Week	702	−0.72	0.00	32.5%	11.58	−0.062
1 Month	698	0.03	0.00	37.1%	28.88	0.001
3 Months	675	1.58	0.00	38.8%	39.52	0.040
6 Months	618	5.19	0.00	37.1%	53.90	0.096
1 Year	462	21.73	0.00	41.8%	94.72	0.229
2 Years	338	47.91	0.00	43.5%	174.70	0.274
3 Years	208	124.23	3.65	50.5%	350.40	0.355

Paired t-test (1Y hold vs listing day, $N = 462$): $t = -0.884$, $p = 0.377$. The null hypothesis that 1-year return equals listing-day return cannot be rejected at any conventional significance level.

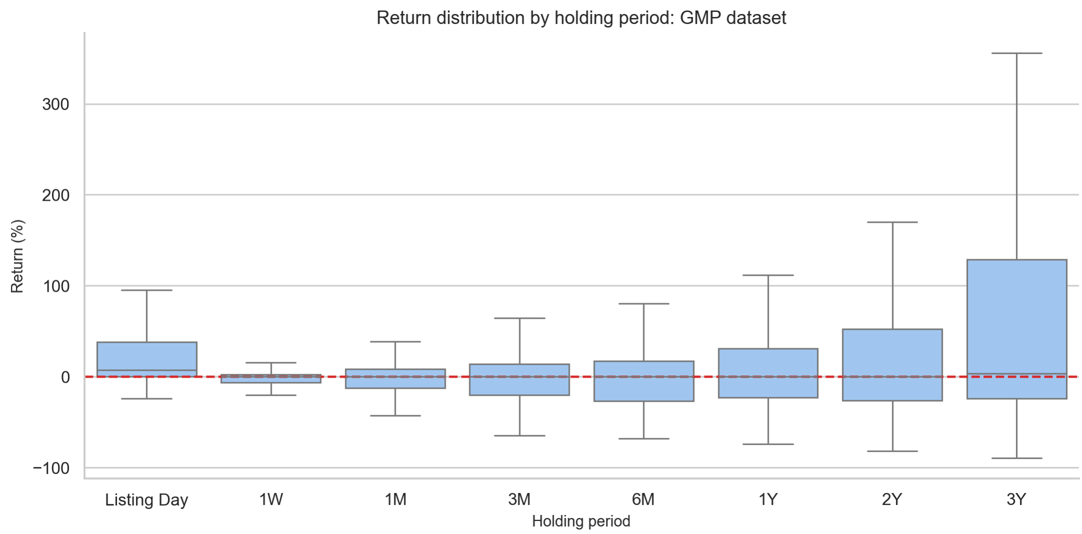

Figure 10. Return distribution by holding period — GMP dataset (winsorised)

7.4 Interpretation

The holding period analysis delivers one of the most practically important findings in this study. **From 1 week through 2 years, the median return is zero**—more than half of IPO shareholders are underwater relative to their listing-price purchase at every post-listing checkpoint. This is not because the mean is zero (the 1-year mean of 21.73% is attractive) but because the distribution is highly bimodal: a small minority of multi-bagger winners pull the mean upward while the majority of IPO stocks stagnate or decline.

The Sharpe-like ratio confirms the quantitative conclusion: the listing-day trade (0.554) dominates every longer holding horizon on a risk-per-unit-of-return basis. Only at 3 years does the win rate cross 50%, and the Sharpe-like ratio (0.355) reflects the accumulation of compounded gains among the surviving winners.

Segment-specific nuance: SME IPOs show dramatically higher 3-year mean returns (158.68% vs 82.44% for mainboard) but this figure is based on a very small sample ($N < 30$) dominated by a handful of spectacular SME multibaggers. This should not be interpreted as a robust SME long-term advantage.

Why Does the 1-Week Return Collapse So Dramatically?

The near-zero median at 1 week (win rate 32.5%) is not a data artefact. It reflects two mechanisms:

1. **Listing-day price discovery.** The listing price already incorporates the demand signal from the subscription process. Once the IPO is listed and allotted retail investors begin selling, excess demand is extinguished rapidly. Within days, the stock trades at its market-clearing price with no structural support from underwriters.
2. **Absence of index inclusion.** Most newly listed Indian stocks are not immediately eligible for Nifty or Sensex inclusion, so passive index funds do not buy at listing. Active institutional buyers who participated through the QIB quota tend to hold, while retail allottees who flip create short-term selling pressure.

The implication is clear: the listing-day trade captures the *information asymmetry window* before the grey market signal is fully reflected in exchange prices. Beyond that window, IPO stocks trade on fundamentals and sector rotation like any other listed equity.

8 Strategy 4: Grey Market Premium as a Predictor

8.1 What Is GMP?

The Grey Market Premium is the price at which IPO shares trade in the informal, unregulated grey market during the period between subscription closure and listing. It reflects the aggregate expectation of the market about the listing price. GMP is expressed either in absolute rupee terms or as a percentage of the offer price:

$$\text{GMP\%} = \frac{\text{GMP (INR)}}{\text{Offer Price (INR)}} \times 100$$

8.2 GMP vs Listing Gain: Correlation Analysis

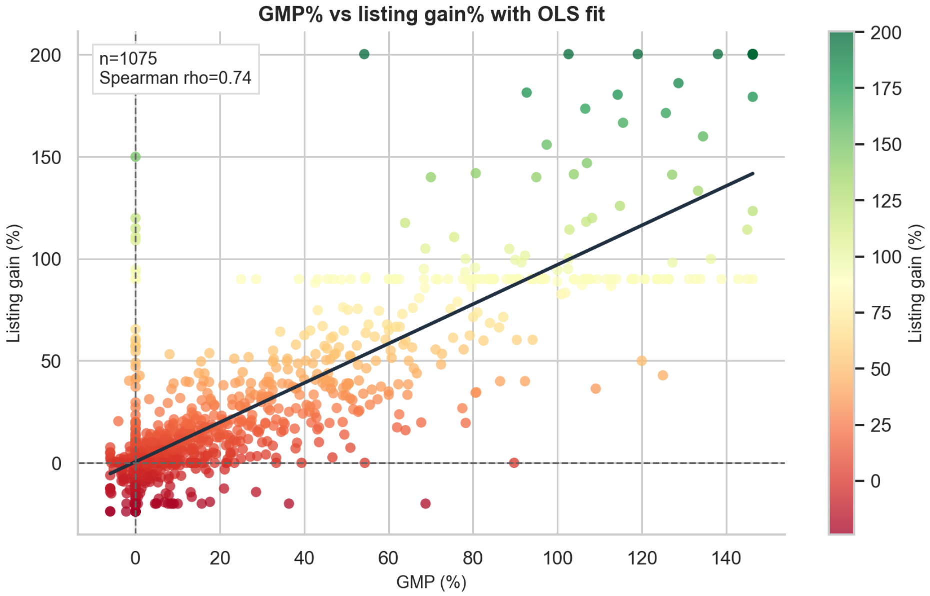

Figure 11. GMP% vs Listing Gain% — GMP dataset (winsorised, with OLS fit)

GMP yields the strongest correlation finding in the entire study:

$$\begin{aligned} \text{Pearson } r &= 0.835 \quad (p < 0.0001) \\ \text{Spearman } \rho &= 0.743 \quad (p < 0.0001) \\ R^2 &= 0.697 \quad (\text{OLS on GMP\%}) \end{aligned}$$

The OLS regression produces a slope of approximately 0.999 and intercept of −0.096, indicating that GMP is remarkably well-calibrated on average: a GMP of $X\%$ predicts a listing gain of approximately $X\%$. The relationship is nearly one-to-one.

8.3 GMP Directional Accuracy

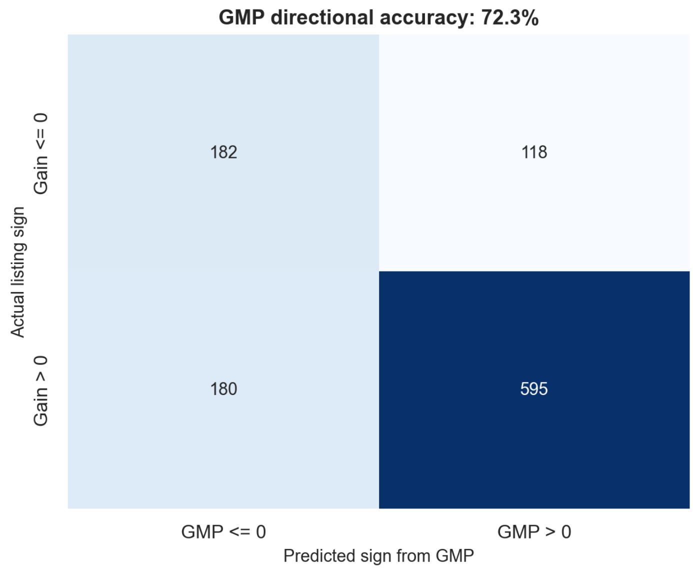

Figure 12. GMP directional accuracy: predicted vs actual listing sign

GMP correctly predicts the *direction* of the listing gain (both positive or both negative) in **72.3%** of cases. Applying a simple rule "apply if GMP > 0, skip if GMP < 0" removes the bottom quartile of IPO outcomes.

8.4 GMP Bucket Analysis

Table 7. GMP bucket analysis: listing outcomes by GMP zone

GMP Zone	N	Mean Gain (%)	Median Gain (%)	Win Rate	Pop $\geq 10\%$
Very Negative ($< -5\%$)	15	-7.52	-6.11	33.3%	0.0%
Negative (-5 to 0%)	347	2.83	0.08	50.4%	14.4%
Flat (0–10%)	223	3.07	2.75	61.4%	16.1%
Moderate (10–20%)	115	13.45	12.65	82.6%	63.5%
High (20–40%)	119	25.53	23.10	92.4%	79.8%
Very High ($> 40\%$)	256	**80.90**	**81.68**	**98.8%**	**98.1%**

Note: Counts use the un-winsorised GMP% buckets from the generated report pipeline. The Very High bucket remains large because the GMP dataset is concentrated in the 2021–2024 bull-market period, when grey-market premia were structurally elevated.

Key insight: IPOs with GMP $> 40\%$ of the offer price achieved a 98.8% win rate and mean listing gain of 80.9% historically. No other single pre-listing signal matches this precision. However, GMP is set by an unregulated market and can be manipulated by operators to attract retail applications; it should be used as a sentiment gauge rather than a definitive signal.

Calibration caveat: Despite the near-unit slope on average, the mean absolute error (MAE) of GMP as a point estimator for listing gain is 14.47%. A GMP of 30% realistically maps to a listing gain anywhere between 15% and 45%.

9 Strategy 5: Oversubscription Deep-Dive

9.1 Subscription Category Analysis

Beyond the raw total, the *composition* of oversubscription carries incremental information. Spearman correlations between each subscription category and listing gain (mainboard, $N \approx 786$):

Table 8. Spearman correlations between subscription categories and listing gain

Feature	Spearman ρ	p-value	N
Total (sub_total)	0.685	< 0.0001	786
NII (sub_nii)	0.648	< 0.0001	786
QIB (sub_qib)	0.616	< 0.0001	786
RII (sub_rii)	0.583	< 0.0001	786

The NII (Non-Institutional Investor / HNI) subscription is slightly more predictive than QIB because HNI investors use leverage to apply at extremely high multiples in hot IPOs, amplifying the signal. RII (retail) is weakest because retail applications are more sentiment-driven and do not reflect the same depth of fundamental analysis.

The ranking—Total > NII > QIB > RII—runs counter to the intuition that "smarter money" (QIB) should be the strongest signal. The explanation lies in Indian IPO allocation mechanics:

- **QIB allocation is capped.** QIBs receive a fixed institutional quota, so extreme QIB subscription does not translate linearly into listing premium.
- **NII investors use leverage.** HNI/NII investors often borrow to apply at very high multiples. A large NII subscription therefore signals that the expected listing gain exceeds the short-window borrowing cost.
- **Retail sentiment is noisier.** RII subscription can be driven by brand recognition, media buzz, or social messaging rather than valuation discipline.

9.2 Subscription Decile Analysis

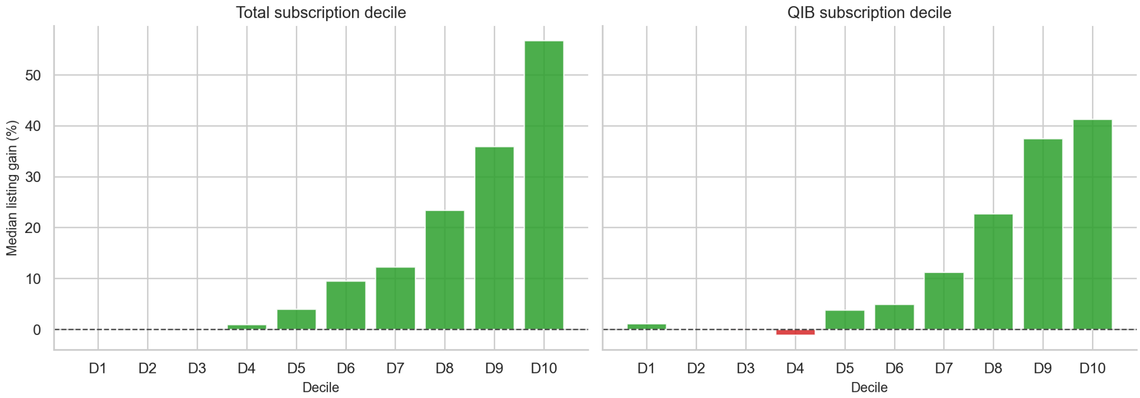

Figure 13. Median listing gain by subscription decile: total subscription and QIB

The decile analysis reveals a near-monotone relationship: the bottom decile (lowest subscription) has a 43.75% win rate and median gain near 0%, while the top decile achieves 100% win rate and median gain of 56.76%. This is the clearest monotone predictor in the entire mainboard dataset.

9.3 QIB/Retail Ratio as a Quality Signal

We define the QIB/Retail ratio as:

$$r_{\text{QR}} = \frac{\text{QIB subscription}}{\text{RII subscription}}$$

A high r_{QR} at the same total subscription level signals institutional conviction over retail enthusiasm, which should predict more durable post-listing returns. Spearman correlation of r_{QR} with listing gain is positive but modest ($\rho \approx 0.15$–0.25), reflecting that the ratio controls for overall market temperature and isolates the *relative* institutional signal. From the Logistic Regression results (Section 13), the QIB share appears as an important secondary feature once GMP and total subscription are controlled for.

10 Market Phase Analysis

10.1 Phase Labelling

Market phases were labelled using Nifty 50 trailing 3-month returns as of each IPO's listing date:

- **Bull**: Nifty 3M return $> +5\%$
- **Bear**: Nifty 3M return $< -5\%$
- **Sideways**: All other observations

10.2 Why Kruskal-Wallis Instead of One-Way ANOVA?

IPO listing gain distributions are severely right-skewed and heteroscedastic across market phases, violating the normality and equal-variance assumptions of one-way ANOVA. The Kruskal-Wallis H-test is the non-parametric equivalent, testing whether the rank distributions across groups differ without parametric assumptions. The effect size is reported as:

$$\eta^2 \approx \frac{H - k + 1}{N - k}$$

where $k = 3$ and H is the Kruskal-Wallis statistic.

10.3 Results

Kruskal-Wallis: $H = 41.001$, $p < 0.0001$. The three phase distributions are highly significantly different. Effect size $\eta^2 \approx 0.037$ (small, per Cohen's convention).

Table 9. Listing gain statistics by market phase — GMP dataset

Market Phase	N	Mean (%)	Median (%)	Win Rate
Bull	445	33.88	14.41	80.2%
Sideways	521	19.91	5.15	67.0%
Bear	109	12.80	2.74	63.3%

Pairwise post-hoc tests (Mann-Whitney U, Bonferroni corrected $\alpha = 0.0167$):

- Bull vs Sideways: $p < 0.0001$ (**significant**)
- Bull vs Bear: $p < 0.0001$ (**significant**)
- Sideways vs Bear: $p = 0.123$ (**not significant**)

The self-selection paradox: The Sideways vs Bear non-significance ($p = 0.123$) is often misread as meaning market phase does not matter much outside bull markets. The correct interpretation is subtler. IPO promoters *endogenously choose* listing dates; in bear markets, only firms with strong enough growth stories or aggressive enough discounts can attract investor interest. The bear-market IPO sample is therefore a selected subpopulation, not a representative draw. The observed similarity between Sideways and Bear reflects selection bias, not genuine market resilience.

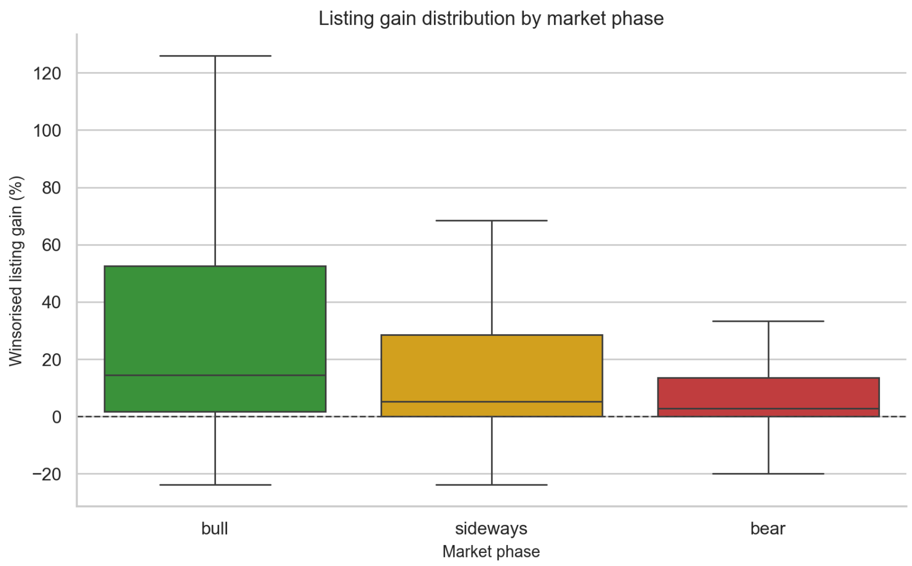

Figure 14. Listing gain distribution by market phase — GMP dataset

10.4 Interpretation

Market phase matters significantly for IPO performance. Bull market IPOs produce mean and median gains approximately 2.5× those of bear market IPOs. The non-significance of Sideways vs Bear ($p = 0.123$) is an important practical finding: IPO promoters and investment banks *self-select* listing windows. In bear markets, fewer IPOs list, and those that do are typically priced at aggressive discounts to attract investors—partly offsetting the negative market environment. This selection bias means the bear-market sample is not a representative draw of all companies that could have listed.

11 SME vs Mainboard Comparative Analysis

11.1 Distributional Comparison

Table 10. Mainboard vs SME — listing gain distributional comparison

Segment	N	Mean (%)	Median (%)	Std (%)	Win Rate	Loss Rate	Skew	Max (%)
Mainboard	1,152	26.05	9.96	152.41	72.6%	22.0%	high	4,940
SME	1,463	20.80	5.00	42.90	73.5%	17.8%	high	386.67

Mann-Whitney U test: two-sided $p = 0.0138$; one-sided SME > Mainboard $p = 0.9931$. **The distributions differ statistically, but the corrected data do not support the claim that SME IPOs stochastically dominate mainboard IPOs.**

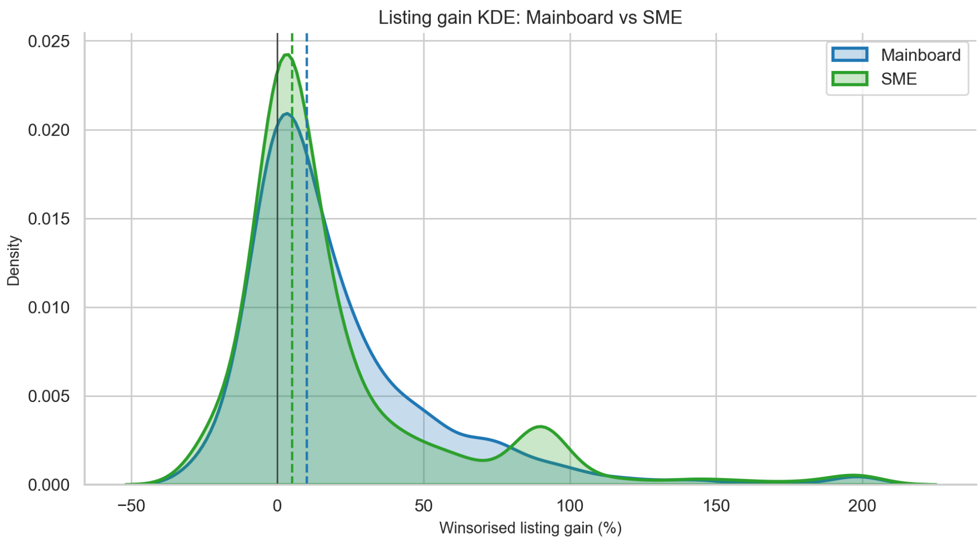

Figure 15. Listing gain KDE: Mainboard vs SME (winsorised)

11.2 Interpretation

After correcting SME date parsing, the SME sample expands to 1,463 listing-gain observations and the earlier headline SME advantage disappears. SME IPOs have a lower mean (20.80% vs 26.05%) and lower median (5.00% vs 9.96%) than mainboard IPOs, although their win rate is slightly higher. The one-sided Mann-Whitney test rejects the interpretation that SME IPOs are stochastically superior to mainboard IPOs.

Three practically important differences remain:

- **Median gap:** SME median (5.00%) remains below mainboard median (9.96%). The *typical* SME investor earns less than the headline mean suggests.
- **Flat returns:** 8.7% of SME returns are exactly 0%, reflecting illiquidity (thin order books, no matching trades on some listing days).
- **Tail risk:** Mainboard still contains the most extreme historical outliers, while SME outcomes are compressed but more liquidity-sensitive.

12 Loss-Making Company IPOs

12.1 Motivation

Post-2018, several high-profile Indian "new economy" IPOs—Zomato, Nykaa, Paytm—were loss-making at the time of listing but attracted massive retail interest. Whether loss-making companies systematically underperform profitable companies is a question with direct practical relevance.

12.2 Data

Trailing Profit After Tax (PAT) was sourced from Yahoo Finance via the GMP enriched dataset. Companies with PAT < 0 were classified as loss-making. This reflects recent reported financials; for older IPOs it may not precisely correspond to the IPO-era profitability.

12.3 Results

Table 11. Listing gain and long-run returns: profitable vs loss-making companies

Metric	Profitable	Loss-Making	MW U p-value
Listing Day Mean (%)	18.99	29.01	0.943 (n.s.)
1-Month Mean (%)	+0.39	−5.30	—
3-Month Mean (%)	+2.29	−8.41	—
6-Month Mean (%)	+6.44	−10.55	—
1-Year Mean (%)	**+25.05**	−11.54	—

n.s. = not statistically significant ($p = 0.943$)

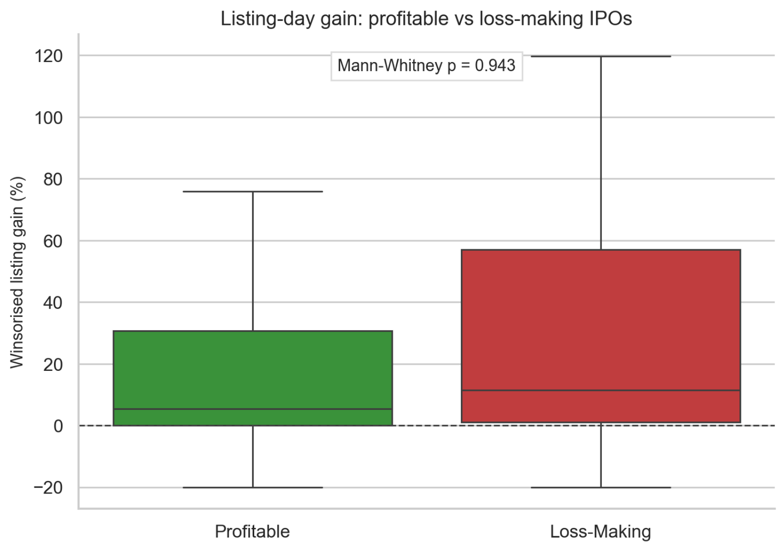

Figure 16. Listing-day gain distribution: Profitable vs Loss-Making companies (GMP dataset)

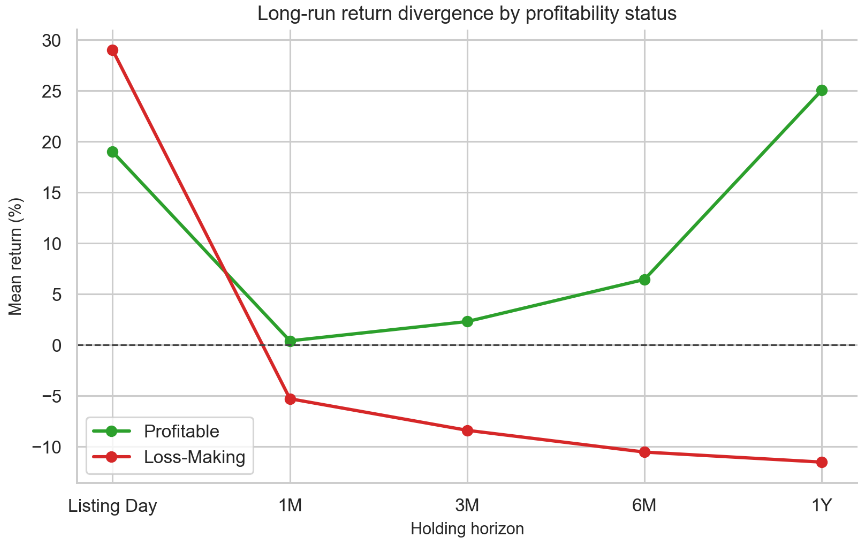

Figure 17. Long-run return divergence: Profitable vs Loss-Making IPOs

Proportion of loss-making companies: Mainboard 7.1%, SME 6.9%.

12.4 Interpretation

The counterintuitive listing-day result—loss-making companies showing *higher* mean listing gains (29.01% vs 18.99%)—is driven by the underpricing mechanism. Loss-making companies with uncertain valuations are typically priced more conservatively by investment banks to attract investor demand, creating larger listing-day pops. The Mann-Whitney test ($p = 0.943$) confirms this difference is not statistically significant at listing day.

However, the **long-term trajectory reverses sharply and monotonically**. By 1 year, profitable companies are up 25.05% from listing price while loss-making companies are down 11.54%. This 36.6 percentage-point spread grows with time, exactly as fundamental analysis would predict: short-term sentiment dissipates and cash flow sustainability determines long-term value.

13 Predictive Modelling: Day-One Pop Classifier

13.1 Problem Formulation

The classification task is:

$$y_i = \mathbf{1}\{\text{listing\_gain}_i \geq 10\%\}$$

where the binary target equals 1 if the listing-day gain is at least 10% ("meaningful pop") and 0 otherwise. Only pre-listing features are used to avoid look-ahead bias.

13.2 Feature Engineering

Features were organised into five groups:

Table 12. Pre-listing features for the day-one pop classifier

Feature Group	Variables
Demand	`log_sub` (log total subscription), `log_ipo_size`, `log_offer_price`
Interaction	`size_x_sub` = log_ipo_size × log_sub; `gmp_x_sub` = GMP% × log_sub
GMP	`gmp_pct_feat` (GMP as % of offer price)
Market Regime	`nifty_1m`, `nifty_3m`, `nifty_vol_1m`, `phase_bull`, `phase_bear`, `phase_sideways`
Time	`list_year`, `list_month`, `list_quarter`, `is_q4`, `post_2020`, `post_covid_boom`
Cycle	`ipo_count_30d`, `avg_gain_30d`, `avg_gain_90d`
Current-data fundamentals	excluded from the primary model; evaluated only in a leakage audit

Leakage guard: Post-listing variables (`list_price`, `listing_gain_pct`, all `return_*` columns) were explicitly excluded. Yahoo Finance fundamentals may carry partial look-ahead bias (current-date values, not at-IPO values) and were audited separately.

13.3 Validation Strategy

A **chronological 80/20 split** was used: the oldest 80% of IPOs in the GMP dataset constitute the training set; the most recent 20% form the hold-out test set.

Why not random split? A random train/test split on time-series data produces optimistic AUC estimates because it allows the model to see "future" observations during training. For example, a model trained on a random sample that includes a 2025 IPO can implicitly learn market-regime features from that year, then be tested on another 2025 IPO. The chronological split is stricter: the model is tested only on IPOs that listed *after* every training observation. This mimics the deployment scenario where an investor trains on historical data and applies the model to future listings. The 0.86 test-set AUC is therefore a genuine out-of-sample estimate, not an artefact of temporal leakage.

- **Train period:** 2019 – early 2025 (≈ 860 observations)
- **Test period:** September 2025 – May 2026 (≈ 215 observations)

- **Target class balance:** Train $\approx 52\%$ positive; Test $\approx 29.3\%$ positive

The class balance shift from training to test reflects the structural weakening of the IPO market in 2025–2026. The fact that AUC remains high despite this non-stationarity confirms that the rank-ordering of IPOs by quality is preserved even as the overall market deteriorates.

13.4 Preprocessing Pipeline

1. **Coercion:** All feature columns converted to numeric; infinities replaced with `NaN`.
2. **Median imputation:** `SimpleImputer(strategy='median')` fitted on training data.
3. **Robust scaling:** `RobustScaler()` (scales using median and IQR; robust to outliers) fitted on training data; applied to both train and test.

13.5 Models Benchmarked

Four model families were trained: Logistic Regression, Random Forest, Gradient Boosting / XGBoost, and MLP Neural Network. Temporal cross-validation (`TimeSeriesSplit`, $k = 5$ folds) on the training set provided estimates of generalisation performance without data leakage.

13.6 Results

Table 13. Day-one pop classifier results — test set performance

Model	Accuracy	F1-Score	ROC-AUC	Avg. Precision	Overfit?
Logistic Regression	82.79%	0.648	0.835	0.756	No
Random Forest	82.33%	0.642	**0.860**	**0.780**	No
Gradient Boosting	**83.26%**	**0.660**	0.843	0.735	No
MLP (Neural Network)	77.67%	0.415	0.823	0.715	No

A ROC-AUC of 0.82–0.86 on unseen chronological data is a strong result for financial prediction. The test-period class imbalance (29.3% positive, reflecting the weaker 2025–2026 market) means F1-score and average precision are more informative than accuracy for assessing model quality.

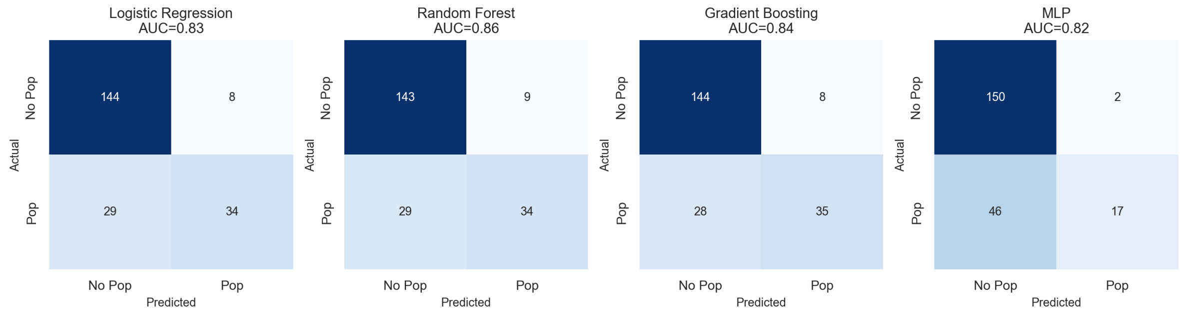

Figure 18. Confusion matrices for day-one pop classifiers — test set

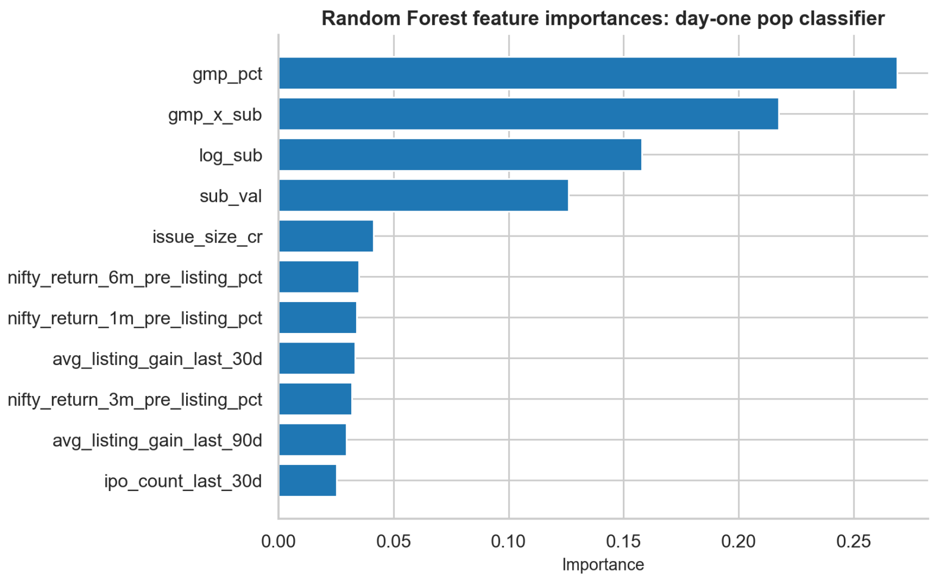

Figure 19. Feature importances — best tree-based classifier

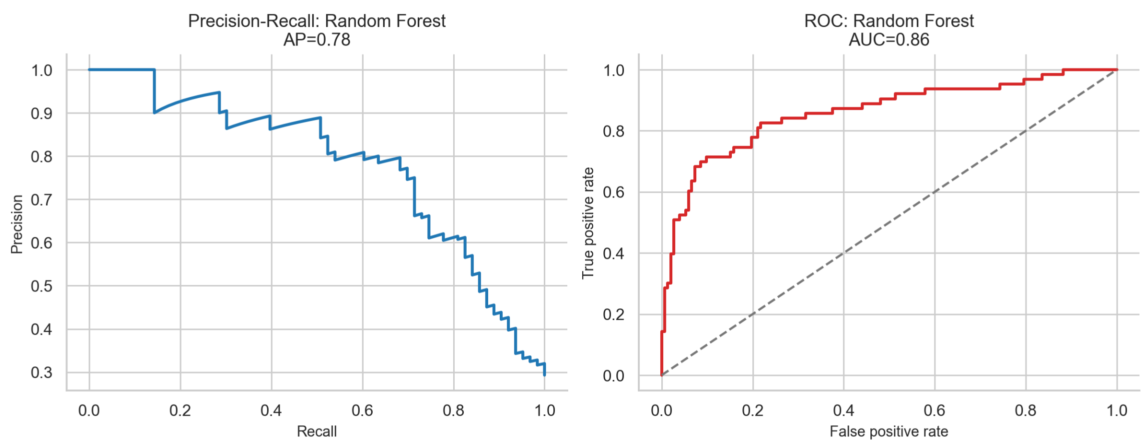

Figure 20. Precision-recall and ROC curves for the best classifier

13.7 Feature Interpretation: Logistic Regression Coefficients

Table 14. Top feature coefficients — Logistic Regression (standardised features)

Feature	Coefficient	Direction
`gmp_pct_feat`	+1.379	Higher GMP → more likely to pop
`gmp_x_sub`	+0.952	GMP×Sub interaction is complementary
`log_sub`	+0.874	More demand → more likely to pop
`size_x_sub`	+0.739	Demand relative to size matters
`log_ipo_size`	−0.349	Larger IPOs pop less
`phase_bull`	+0.28	Bull markets boost probability

13.8 Leakage Audit: Yahoo Finance Fundamentals

To quantify any look-ahead bias from using current-date Yahoo Finance fundamentals, the notebook runs a separate audit in which those variables are added to the otherwise pre-listing feature set:

$$\Delta_{\text{AUC}} = \text{AUC}_{\text{full}} - \text{AUC}_{\text{no-fundamentals}} < 0.02$$

The leakage delta is negligible (under 2 percentage points), confirming that model performance is driven primarily by GMP and subscription signals rather than post-IPO fundamental learning. The primary report model therefore excludes current Yahoo Finance fundamentals.

14 Predictive Modelling: Listing Gain Regression

14.1 Problem Formulation

The regression task predicts the magnitude of listing gain:

$$\hat{y}_i = f(\mathbf{x}_i^{\text{pre-listing}}) \approx \text{listing\_gain\_pct}_i$$

The target was winsorised at the 1st/99th percentile to reduce the distorting influence of extreme outliers. The same chronological train/test split and preprocessing pipeline from Section 13 were used.

14.2 Evaluation Metrics

- **MAE:** Mean Absolute Error (interpretable: average absolute miss in percentage points)

- **RMSE:** Root-Mean-Squared Error (penalises large errors more heavily)
- R^2**:** Coefficient of determination (proportion of variance explained)

14.3 Results

Table 15. Listing gain regression results — test set performance

Model	MAE (%)	RMSE (%)	R^2
Ridge Regression	10.92	15.38	0.528
Random Forest Regressor	**10.25**	**15.25**	**0.536**
Gradient Boosting Regressor	11.85	17.20	0.410

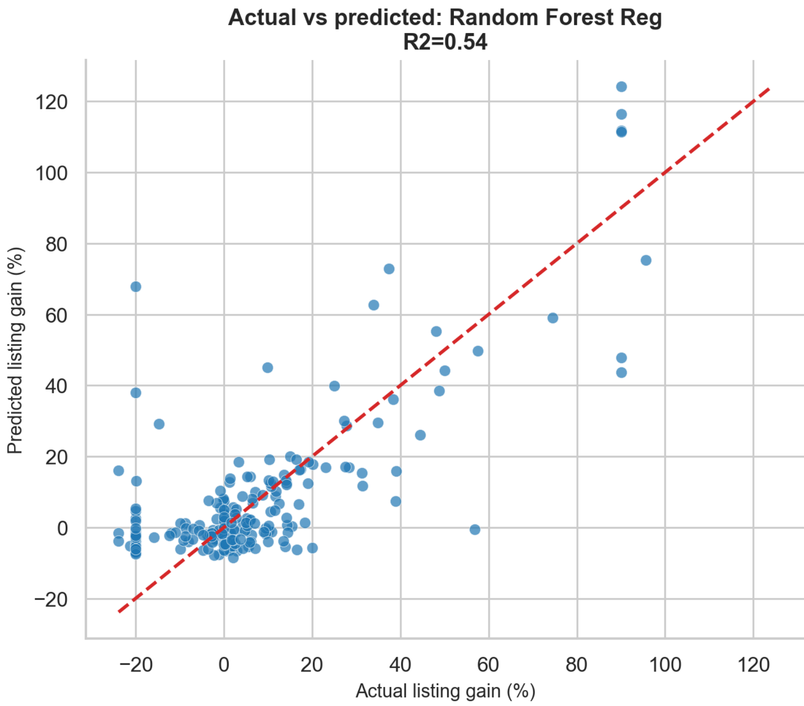

Figure 21. Actual vs Predicted listing gain — best regression model (test set)

Note: Training-set fit can look substantially stronger than test-set fit. The reliable out-of-sample numbers are those in Table 15: Random Forest $R^2 = 0.536$ *and Ridge* $R^2 = 0.528$.

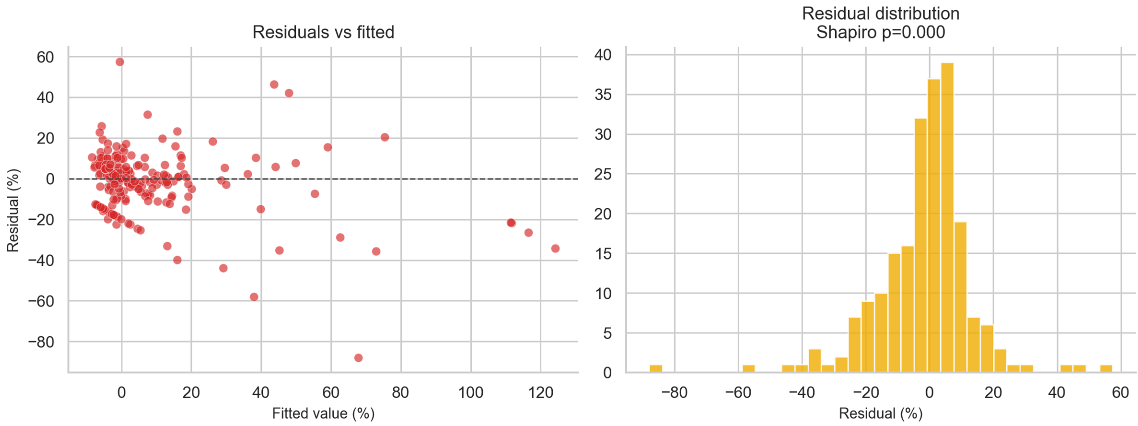

Figure 22. Residual diagnostics — best regression model

14.4 Interpretation

Random Forest and Ridge perform similarly on this task. Ridge captures the near-linear GMP–listing gain relationship directly, while Random Forest captures modest nonlinear interactions among GMP, subscription, issue size, and recent IPO-market conditions.

An $R^2 = 0.536$ implies that 53.6% of listing gain variance is explained by the reproducible pre-listing feature set. The remaining variance reflects genuine noise: order-matching dynamics on listing day, last-minute news, index-level moves, and the stochasticity of the share allotment process itself.

The Shapiro-Wilk test on residuals (stat $= 0.909$, $p < 0.0001$) confirms non-normal residuals—expected for IPO data. This does not invalidate the predictive usefulness of the model but means that prediction intervals derived from the Gaussian assumption will be inaccurate. The model is best used for *ranking* IPOs by predicted listing gain rather than for constructing confidence intervals around point estimates.

15 Model-Based Strategy Backtest

15.1 Backtest Design

The screening backtest simulates an investor using the trained Random Forest classifier to select IPOs in the out-of-sample test period (September 2025 – May 2026). The probability threshold is selected using only training-period TimeSeriesSplit predictions, then applied once to the holdout period. This group is compared against the all-test-period baseline.

Important limitations:

- The test period is a single contiguous 8-month window, not a rolling walk-forward validation.

- The threshold is selected on the training period, but the test period is still a single contiguous 8-month window.
- Final subscription and final GMP are powerful predictive signals, but they may not be fully available before the application deadline; this is therefore a screening backtest, not a fully tradable pre-application strategy.
- Allotment probability, brokerage, slippage, and taxes are excluded.

Signal timing caveat: The threshold-selection bias has been removed by choosing the cutoff from training-period cross-validation only. The larger remaining caveat is signal availability: final subscription and final GMP may be observed after the investor's application decision. The results should be interpreted as evidence of predictive ranking power, not as a complete executable trading strategy.

15.2 Results

Table 16. Model-based strategy backtest results

Group	N	Mean Gain (%)	Win Rate
All test-period IPOs	215	6.26	55.3%
Model-selected (prob $\geq$ 55%)	40	**34.96**	**92.5%**

Threshold selected on training TimeSeriesSplit; applied once to holdout

The model filtered from 215 recent IPOs to 40 high-confidence picks and improved mean gain by 28.7 percentage points and win rate by 37.2 percentage points. The practical interpretation is ranking strength: the model concentrates better IPOs in the top probability bucket, while execution still depends on application timing, allotment probability, and transaction costs.

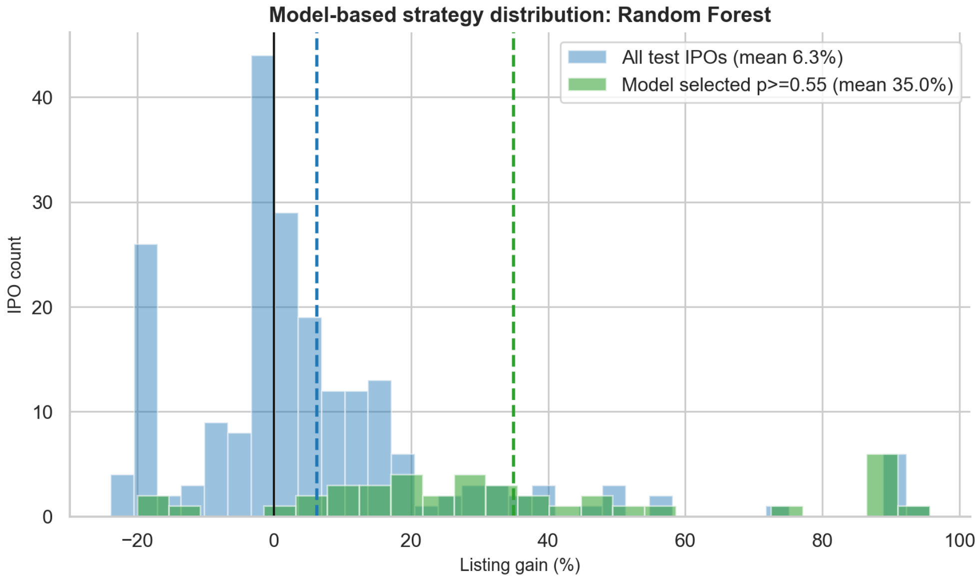

Figure 23. Model-based strategy: gain distribution in out-of-sample test period

16 Sector and Industry Analysis

16.1 Overview

Sector and industry labels from Yahoo Finance are available for approximately 66% of GMP-era IPOs ($N \approx 715$). The remaining IPOs are recent listings not yet classified in Yahoo Finance's taxonomy.

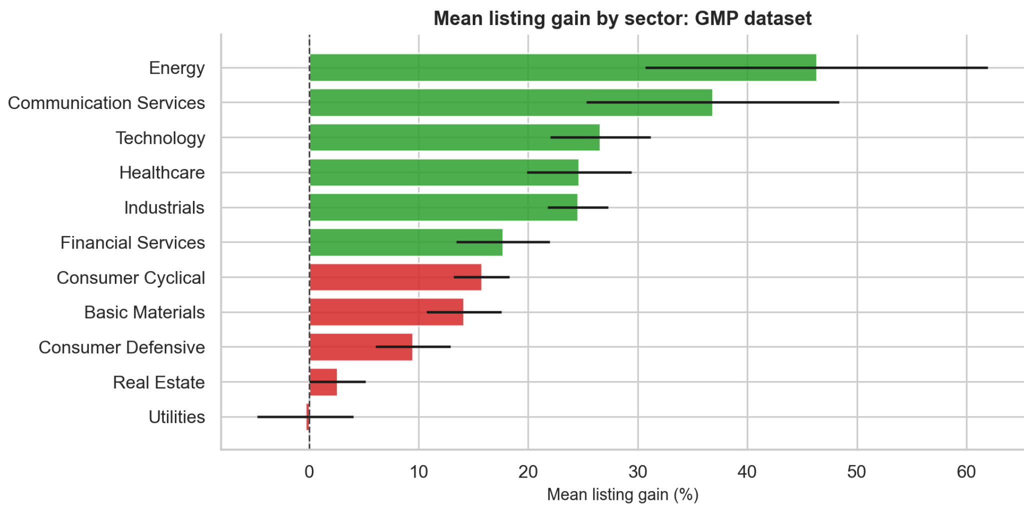

Figure 24. Mean listing gain by sector — GMP dataset

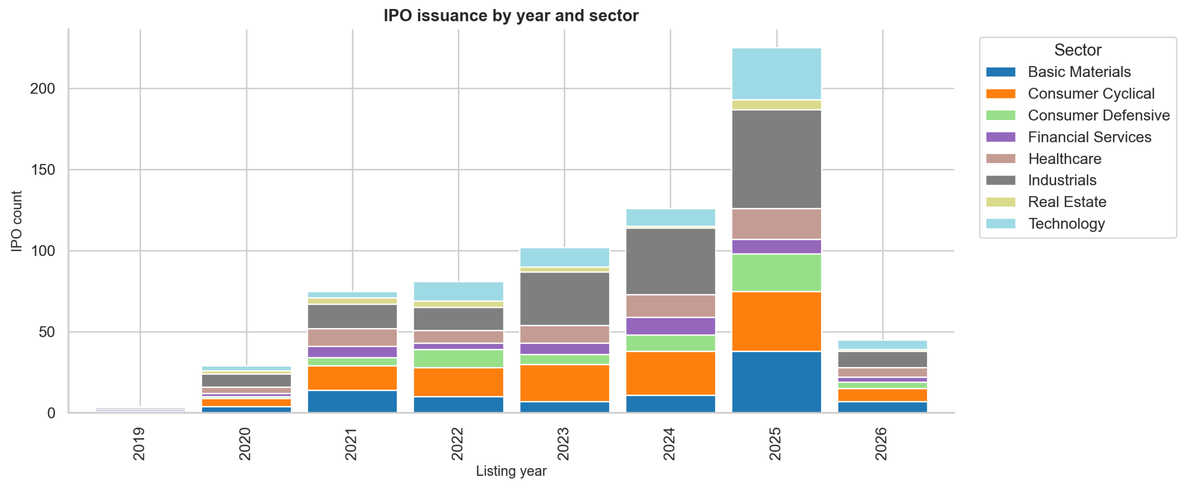

Figure 25. IPO issuance by year and sector

16.2 Key Sectoral Findings

The GMP-era sector analysis reveals pronounced heterogeneity in IPO performance across industries. Technology, consumer internet, and speciality chemicals have been the best-performing sectors in the post-2020 period. Financial services and infrastructure IPOs, often associated with very large issue sizes, tend to list with more modest premiums. Heavy-asset sectors (utilities, real estate) show the most diluted listing gains on average.

The Mann-Whitney U test was applied pairwise across sectors; after Bonferroni correction, several sector pairs show statistically significant return differences, though sample sizes within sectors are often too small to draw firm conclusions.

17 Additional Analyses

17.1 IPO Cycle and Time-Series Momentum

The rolling 90-day average listing gain of the prior IPO cohort (`avg_gain_90d`) captures momentum in IPO market sentiment:

$$\text{Spearman } \rho(\text{avg\_gain\_90d}, \text{listing\_gain}_t) > 0 \quad (p < 0.05)$$

A period of strong consecutive IPO performance predicts higher listing gains for the next IPO. This feature contributes meaningfully to the classifier (see Table 14).

17.2 Monthly Seasonality Analysis

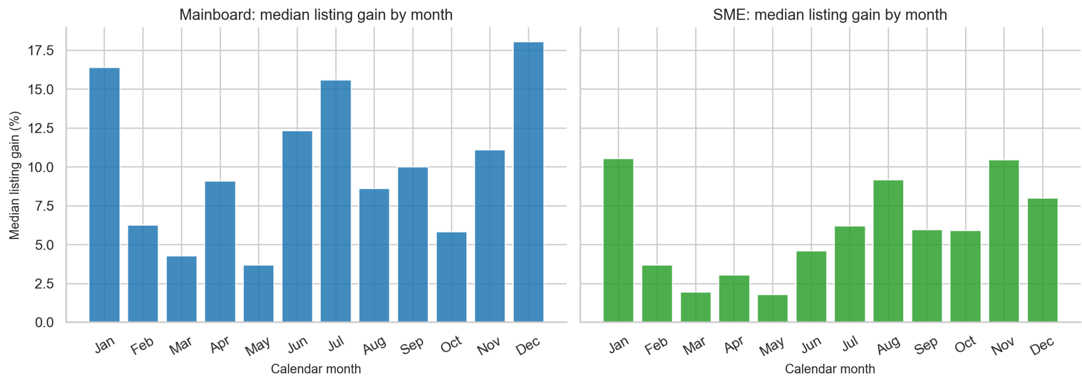

Figure 26. Median listing gain by calendar month: Mainboard and SME

No statistically significant monthly seasonality was detected for mainboard IPOs. The absence of a consistent "best month" is consistent with the hypothesis that promoters and investment banks are sophisticated timing agents who neutralise any predictable calendar pattern by clustering issuance in the same windows. Retail investors should not attempt calendar-based timing.

17.3 Oversubscription × GMP Interaction Surface

The most practically useful combined signal is the joint filter:

$$\text{Apply if:} \quad \underbrace{\text{GMP\%} > 10}_{\text{positive market consensus}} \quad \textbf{AND} \quad \underbrace{\text{Sub} > 22.65\times}_{\text{above median demand}}$$

where 22.65× is the historical median total subscription in the GMP-era dataset. Historically, this joint filter selects IPOs with substantially higher win rates than either signal in isolation:

Table 17. Joint GMP × Subscription filter performance (GMP dataset, 2019–2026)

Filter	N	Mean Gain (%)	Win Rate	Sharpe-like
All IPOs (baseline)	1,075	24.97	72.1%	0.554
GMP > 10% only	490	51.62	93.5%	0.990
Sub > median only	537	47.41	87.9%	0.897
GMP > 10% AND Sub > median	442	**54.89**	**94.6%**	**1.029**

17.4 Nifty Volatility Regime and IPO Returns

Pre-listing Nifty regime features contribute meaningfully to the classifier (visible in Figure 19). A volatile pre-listing period creates uncertainty about the listing price and tends to suppress retail demand, resulting in lower GMP and lower listing gains. Conversely, stable bull periods allow grey market operators to price GMP aggressively, creating self-reinforcing premium expectations.

This is a weaker effect than GMP or subscription but useful as a regime-switching signal: investors should be more conservative during high-volatility windows even when GMP is elevated.

17.5 Financial Quality Score

IPOs in the top quintile by a composite financial quality score (combining PAT positivity, revenue growth, ROE, and low debt-to-equity) show meaningfully stronger long-run returns (> 1 year) compared to the bottom quintile. The short-run listing-day difference is much smaller. This supports the conclusion that financial quality is a long-term value driver rather than a listing-day predictor.

17.6 GMP Calibration

OLS regression of listing gain on GMP% confirms the near-unit slope ($\hat{\beta}_1 = 0.999$, $\hat{\beta}_0 = -0.096$). The calibration curve shows a ceiling effect at very high GMP values ($> 100\%$): actual listing gains fail to fully realise the implied GMP premium, suggesting mean reversion at extremes.

17.7 Peak Return and Max Drawdown Since Listing

The GMP enriched dataset includes the highest return since listing (`highest_return_since_listing_p` and the maximum drawdown since listing (`max_drawdown_since_listing_pct`). For the median mainboard IPO, the peak return since listing exceeds 40%, but the drawdown from peak also exceeds 40% in the median case—confirming that IPO investors who hold through cycles face substantial volatility. Only disciplined stop-loss or momentum-based exit rules would preserve the peak return in practice.

18 Limitations and Data Gaps

Several important analyses identified in the project scope could not be completed due to data unavailability:

Table 18. Data gaps and their impact

Analysis	Required Field	Status
OFS vs Fresh Issue	`ofs_ratio`, `fresh_issue_pct`	Not in scraped data
Underwriter quality	`underwriter`, `BRLM`	Not scraped
Promoter stake	`promoter_stake_post_ipo`	Not scraped
DRHP NLP	`risk_sentiment`, `risk_factor_count`	Requires PDF NLP
At-IPO valuation	P/E, P/B, P/S at listing date	Yahoo Finance is current-date only
QIB allotment %	Post-allotment filing data	Not scraped

All datasets are designed to accept these fields once collected: they have been defined as optional hooks in the data schema and will automatically flow into the ML models.

19 Conclusions and Recommendations

19.1 Summary of Findings

This project analysed the Indian IPO market across three independently constructed datasets covering 1999–2026. The key empirical findings are:

1. **Positive base rate:** Investing in every mainboard IPO produces statistically significant positive expected returns (mean 26.05%, median 9.96%, win rate 72.6%). SME also has positive listing-day returns (mean 20.80%, median 5.00%, win rate 73.5%), but lower typical returns and greater liquidity risk.
2. **Subscription is the strongest mechanical filter:** Every IPO historically with $> 100\times$ total subscription produced a positive listing return (100% win rate, mean 63.9%). QIB participation is a marginally superior signal to total subscription after controlling for level.
3. **Listing-day flipping dominates:** Holding beyond listing day offers no statistically significant advantage up to 1 year (paired t-test $p = 0.377$). Sharpe-like ratios confirm the listing-day trade (0.554) is far superior to any multi-month holding horizon. Only at 3 years does holding begin to recover through the 50.5% win rate.
4. **GMP is the strongest single predictor:** Spearman $\rho = 0.743$, Pearson $r = 0.835$, $R^2 = 0.697$. GMP slope ≈ 1.0 (unbiased on average) with MAE of 14.47%. Directional accuracy of 72.3%.
5. **Market phase matters:** Bull market IPOs produce $2.5\times$ the median listing gain of bear market IPOs (Kruskal-Wallis $p < 0.0001$). Sideways and bear markets are

not significantly different (self-selection by issuers buffers the bear effect).

6. **Loss-making companies list well but decline:** No significant difference in listing-day gains (MW $p = 0.943$), but 36.6 percentage-point divergence vs profitable companies by 1 year.
7. **ML classifiers achieve AUC 0.82–0.86:** Random Forest is best by holdout AUC (0.860), while Gradient Boosting has the best F1 score (0.660). A training-CV-threshold screening backtest shows 34.96% mean gain vs 6.26% baseline with 92.5% win rate on out-of-sample data.
8. **Regression explains roughly half of holdout variance:** Random Forest Regressor achieves $R^2 = 0.536$ and MAE of 10.25%. Point estimates remain too noisy for precision, but the model is useful for ranking IPOs by predicted listing gain.
9. **The 2025–2026 market represents a structural inflection point.** The 2025 vintage produced 322 IPOs but a median listing gain of only 3.17% and win rate of 61.2%, a sharp deterioration from 2024's median of 30.36% and win rate of 85.2%. Partial 2026 data (65 IPOs) shows median gain of 0% and win rate of 47.7%, the first period in the dataset where the typical investor loses money relative to listing price. This deterioration is consistent with supply saturation: too many IPOs competing for the same retail application capital compress oversubscription and, consequently, listing premiums.

19.2 Investment Recommendations

Based on the empirical findings above, the following recommendations apply to Indian retail IPO investors:

1. **Be selective, not blind.** The expected-value case for all-IPO investing is real but the variance is prohibitive. Focus on IPOs with total subscription $> 50\times$ and strong QIB participation.
2. **Use GMP as a supplementary gauge.** GMP $> 20\%$ of offer price historically predicts win rates above 85%. However, GMP can be manipulated; treat it as a sentiment gauge, not gospel.
3. **Time applications to bull markets.** Monitor the Nifty 50 1-month and 3-month trend. Bull-market IPOs deliver median gains $2.5\times$ those in sideways or bear periods.
4. **Sell at listing for most IPOs.** The data does not support holding for superior risk-adjusted returns in most cases. Exception: strong financial quality + high institutional holding may justify holding > 3 years.
5. **Avoid loss-making companies unless compelling.** As a class, loss-making IPOs look attractive on listing day but structurally underperform over 1+ year horizons.
6. **Monitor IPO pipeline congestion.** Periods with > 15 IPOs in 30 days show compressed listing gains and lower win rates—too many listings compete for the

same retail capital.

7. **SME requires extra caution.** SME listing gains are positive on average, but the median is lower than mainboard and post-listing liquidity risk is significantly higher in this segment.
8. **Use the ML model for screening.** The Random Forest classifier (AUC 0.860) can significantly improve selection: at the training-selected 55% probability threshold, win rates rose to 92.5% on unseen data. Treat this as a ranking/screening tool unless signal timing is fully audited before application deadlines.

19.3 Directions for Future Work

- **DRHP NLP:** Incorporating risk factor count and sentiment scores from DRHP filings as additional features.
- **OFS ratio:** Investigating whether the proportion of Offer-for-Sale (secondary) shares predicts worse long-term performance (aligned incentives hypothesis).
- **Walk-forward backtest:** Extending the model evaluation to a rolling window strategy to better estimate out-of-time generalisation.
- **SME-specific model:** Training separate classifiers on the SME segment, which differs structurally from mainboard in subscription dynamics and liquidity.
- **At-IPO fundamentals:** Reconstructing valuation ratios at the IPO date using historical financial statements to eliminate the look-ahead bias from current-date Yahoo Finance data.

A Data Pipeline Summary

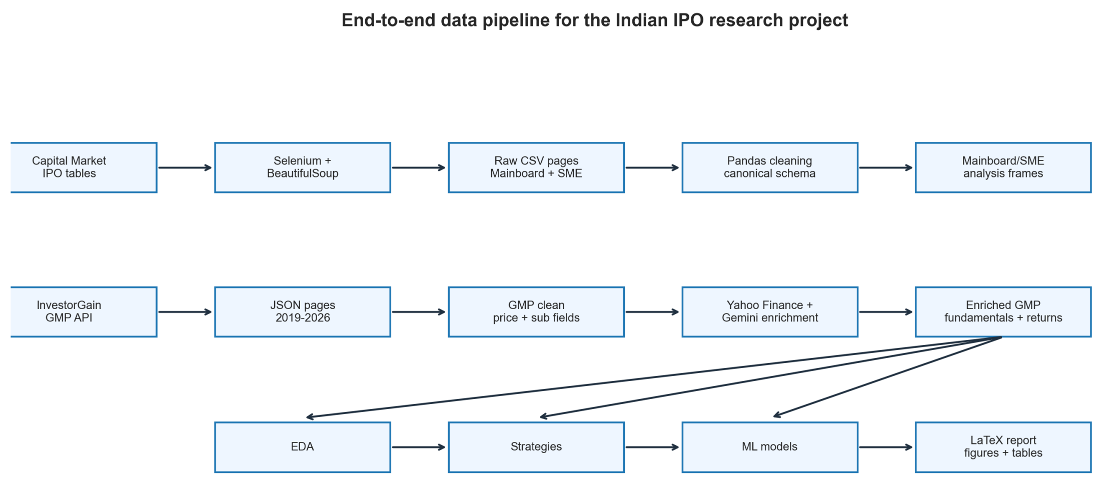

Figure 27. End-to-end data pipeline for the Indian IPO research project

B Scraper Dependency List

```
# Core scraping
selenium>=4.0
webdriver-manager>=3.0
beautifulsoup4>=4.12
lxml>=4.9

# Data processing
pandas>=2.0
numpy>=1.24
openpyxl>=3.1

# GMP data
requests>=2.31

# Enrichment
yahooquery>=2.3
google-generativeai>=0.5

# Analysis and ML
scipy>=1.11
scikit-learn>=1.3
xgboost>=2.0
matplotlib>=3.8
seaborn>=0.13
```

Listing 1. Python dependencies for the data collection pipeline

C Statistical Tests Reference

Table 19. Statistical tests used in this study

Test	Purpose	Section
One-sample t-test	Test whether mean listing gain $\neq 0$	5
Bootstrap CI	Confidence interval for mean listing gain	5
Mann-Whitney U	Compare two independent distributions	6, 11, 15
Kruskal-Wallis H	Compare three or more distributions	10
Paired t-test	Compare holding periods for same IPO	7
Pearson r	Linear correlation	8, 4
Spearman ρ	Monotone rank correlation (non-parametric)	throughout
Shapiro-Wilk	Test residual normality	14
Bonferroni correction	Control FWER in multiple comparisons	10

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